

Hanane Esfandiar

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Education

Ferdowsi University of Mashhad <i>BSc in Computer Science</i> • CGPA: 3.69 (17.52/20) • Last two years GPA: 3.72 (17.85/20)	Mashhad, Iran Sep 2019–Sep 2024
Farzanegan High School (NODET) <i>Diploma in Mathematics and Physics</i> • CGPA: 19.72/20	Babol, Iran Sep 2016–Sep 2019

Research Interests

• Natural Language Processing (NLP) • AI Safety, Ethics, and Alignment	• Multi-Agent Systems • AI and Human-Computer Interaction (HCI)
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Research Experience

Research Assistant <i>Sharif University of Technology, Tehran, Iran</i> • Conducted literature review of educational AI systems. • Identified gaps in LLM-based tutoring systems' cognitive detection. • Proposed Retrieval-Augmented Generation(RAG) to reduce hallucinations. • Proposed framework for adaptive responses based on student performance.	Jun 2024–Oct 2024
Research Assistant <i>Ferdowsi University, Mashhad, Iran</i> • Evaluated transformer architectures for biomedical paper summarization • Compared extractive vs abstractive summarization with custom preprocessing. • Fine-tuned Longformer and BigBird-PEGASUS models using PyTorch. • Earned thesis grade of 19/20 with faculty recognition.	Jan 2024–Sep 2024

Teaching Experience

Teaching Assistant, Ferdowsi University of Mashhad • Data Mining (Head TA) • Artificial Intelligence • Algorithm Design and Analysis • Advanced Programming • Fundamentals of Programming	Sep 2023–Feb 2024 Feb 2023–Jul 2023 Sep 2022–Feb 2023 Feb 2022–Jul 2022 Sep 2021–Feb 2022
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Professional Skills

• Programming Languages: Python, C/C++, Julia • Operating Systems: GNU/Linux (LPIC 1) • Tools/Libraries: $\text{\LaTeX}$, Git, Jupyter Notebook, MATLAB • AI/ML Frameworks: TensorFlow, PyTorch, Scikit-learn • Soft Skills: Teamwork, Communication, Problem-Solving, Teaching	
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Work Experience

Technical Writer <i>AI Agent Simplified</i> - Founded <i>AI Agent Simplified</i>, a technical newsletter with 37k subscribers. - Researched and wrote technical content on AI Agents for a niche audience. - Collaborated with subject-matter experts to validate technical details. - Synthesized research papers into accessible newsletter content.	Jun 2024–Present
Research Scientist <i>Part Research Institute, Mashhad, Iran</i> - Conducted literature reviews on federated learning for network data management. - Analyzed privacy-preserving distributed learning methods and identified scalability issues. - Coordinated educational workshop curriculum aligning student skills with industry demands. - Developed programming contest proposals securing funding and stakeholder participation.	Sep 2023–Mar 2024
Network Engineering Intern <i>Part Software Group, Mashhad, Iran</i> - Completed CCNA training through lectures and hands-on theoretical assignments. - Explored AI-driven network optimization including anomaly detection and traffic forecasting. - Delivered standout technical presentation connecting AI with network engineering. - Earned full-time offer as top performer among 25 interns.	May 2023–Jul 2023

Notable Academic Projects

Transformer-based Text Summarization (thesis) Tokenized the dataset using the BART tokenizer with specific truncation and padding strategies. Fine-tuned the pre-trained facebook/bart-large model using a Sequence-to-Sequence (Seq2Seq) objective. Achieved a cosine similarity score of 0.5446, outperforming baseline configurations in semantic performance.	
Artificial Intelligence - Implemented six AI search algorithms (BFS, DFS, IDS, UCS, Best-First, A*) for Sokoban puzzle-solving. - Designed Manhattan distance heuristic to optimize informed search algorithms and reduce exploration space. - Implemented Backtracking with Forward Checking using MRV, LCV, and AC3 constraint propagation techniques.	
Data Mining - Implemented KNN classifier for Persian digit recognition achieving 97%+ accuracy. - Implemented LDA dimensionality reduction on Iris dataset for binary classification. - Analyzed CSV dataset using pandas for data manipulation and matplotlib for horizontal bar chart generation.	

Awards and Honors

Scientific Association of Computer Science at FUM <i>Ferdowsi University, Mashhad, Iran</i> Elected as the Main Member	Sep 2020–Sep 2021
Ranked in Top 1% among 165,000+ candidates Iranian Nationwide University Entrance Examination (Konkoor)	May 2019–Jun 2019
11th International Scientific League of Paya <i>Shahid Beheshti University, Tehran, Iran</i> Winner of the Gold Group Medal, in the Research League	Aug 2018–Oct 2018
Recognized as Exceptionally Talented By National Organization of Developing Exceptional Talents (NODET)	Aug 2016–Sep 2019